Project Provenance: A Graph-Hybrid Stateful Auditor for Long-Form Narratives

April 23, 2026

Abstract

Maintaining logical consistency in long-form narratives remains a significant challenge for human authors and automated systems alike. This paper presents *Project Provenance*, a hybrid architecture that combines Neural Language Processing (NLP), Knowledge Graph (KG) modeling, and Natural Language Inference (NLI) to detect narrative "continuity glitches." By representing the story world as a stateful graph in Neo4j, our system overcomes the context decay limitations of large language models. Experimental results on a seeded six-chapter dataset demonstrate that our Graph-Hybrid approach achieves an 84.6% precision rate, significantly outperforming traditional semantic-only baselines which suffer from a 1.7% precision rate due to linguistic noise.

1 PROJECT IDENTIFICATION

- **Project Title: Project Provenance: A Graph-Hybrid Stateful Auditor for Long-Form Narratives**
- **Team Members:** Aarpeet Chandrasekhar(IMT2023011), Kanav Bhardwaj(IMT2023024), Suryans Dash (IMT2023041), Bhavya Jain(IMT2023050)
- **Technical Framework:** Agentic Graph-Based RAG (Retrieval-Augmented Generation/Detection)
- **Repository:** github.com/Paradox-73/Provenance

2 Introduction

In the domain of long-form storytelling, from epic novels to serialized teleplays, logical consistency is the foundation of the "suspension of disbelief." As narrative complexity grows, authors often fall victim to the "Continuity Glitch"—logical errors where characters appear in two places simultaneously, items vanish without explanation, or physical attributes mutate between chapters. These errors are not merely aesthetic; they degrade the structural integrity of the story and can lead to significant reader frustration.

Current automated tools for narrative analysis, primarily based on Large Language Models (LLMs), struggle with two fundamental issues: **Context Decay** and **Lack of State Memory**. LLMs possess finite context windows and cannot effectively "remember" a minor detail from the first chapter when auditing the fiftieth. Furthermore, semantic models like Natural Language Inference (NLI) evaluate sentences in isolation, lacking a persistent world model to track the evolving state of entities.

Project Provenance addresses these gaps by introducing a Graph-Hybrid (GH) Stateful Auditor. The system ingests raw narrative text, extracts structural facts (triples), and anchors them into a Neo4j Knowledge Graph. This graph serves as an external, non-decaying memory of the story's world state, allowing for deterministic conflict detection through logical queries. This paper provides a deep analysis of the architecture, its performance against semantic baselines, and the implications for the future of narrative forensics.

3 The "Continuity Glitch" Problem Space

Narrative consistency can be categorized into four primary logical domains, each posing unique challenges for automated detection:

1. **Temporal/Spatial Consistency:** Tracking the physical location of entities across time. A common error is "impossible movement," where a character travels between distant locations in zero elapsed time.
2. **Inventory Persistence:** Maintaining the state of possession for unique objects. An item cannot be both possessed by a character and lost in a different location simultaneously.
3. **Identity and Role Stability:** Ensuring that entities maintain their defined roles or professional identities unless a transformation is narratively justified.
4. **Attribute Invariance:** Consistent physical descriptions. A character described as having a "jagged scar" in Chapter 1 should not have a "smooth face" in Chapter 2 without a causal explanation.
5. **Attribute Transfer (Trait Theft):** A specialized category where a unique physical trait assigned to one character (e.g., a specific scar) is later attributed to another character without explanation, suggesting an identity collision.
6. **Environmental and Physical State Paradoxes:** Mutations of the surrounding world or inanimate objects that violate the laws of physics, such as a liquid glass of water becoming "frozen solid" instantly in a room-temperature environment.

Traditional NLP approaches attempt to solve these through sliding-window semantic comparison. However, as our results will show, these models are easily confused by the rich, metaphorical language typical of creative writing.

4 Architecture and Approach

The Provenance architecture is a modular, multi-stage pipeline designed to bridge the gap between creative prose and formal logic.

4.1 Preprocessing and Data Ingestion

To ensure context preservation across document boundaries, the system implements an overlapping chunking strategy. Each chapter is divided into segments with a 20% overlap, ensuring that entities and their relationships are captured even when they span the edges of a chunk.

4.2 Neural Extraction Layer

The core of the extraction layer is a Transformer-based pipeline. We utilize the `spacy en_core_web_trf` model for its high-accuracy NER and dependency parsing. **Coreference Resolution:** One of the most critical components is the `fastcoref` neural coref module. In literature, characters are frequently referred to by pronouns ("he", "she") or descriptions ("the captain"). Without coreference resolution, the system would create redundant nodes for the same entity. **Triple Extraction:** A custom SVO (Subject-Verb-Object) extractor identifies functional relationships. We prioritize predicates that map to our logical ontology: `LOCATED_AT`, `POSSESSES`, `LOST`, and `HAS_ATTR`.

4.3 Knowledge Graph Modeling (The Story Bible)

Extracted facts are ingested into a Neo4j database utilizing a structured ontology known as the *Ship Schema*. **Entity Deduplication:** Every entity is assigned a `canonical_id`. For example, "Kael", "Commander Kael", and "the commander" are all mapped to the same node (`Person:Kael`). **Spatial Awareness:** The graph includes a hierarchy of locations. By injecting a predefined vessel schema (e.g., Decks 1-10, Bridge, Engineering), the system can perform path-finding queries. If the graph shows a character is at `Location:Bridge` and `Location:Engineering` simultaneously, it flags a conflict because these nodes are not hierarchically linked (i.e., Engineering is not *part of* the Bridge).

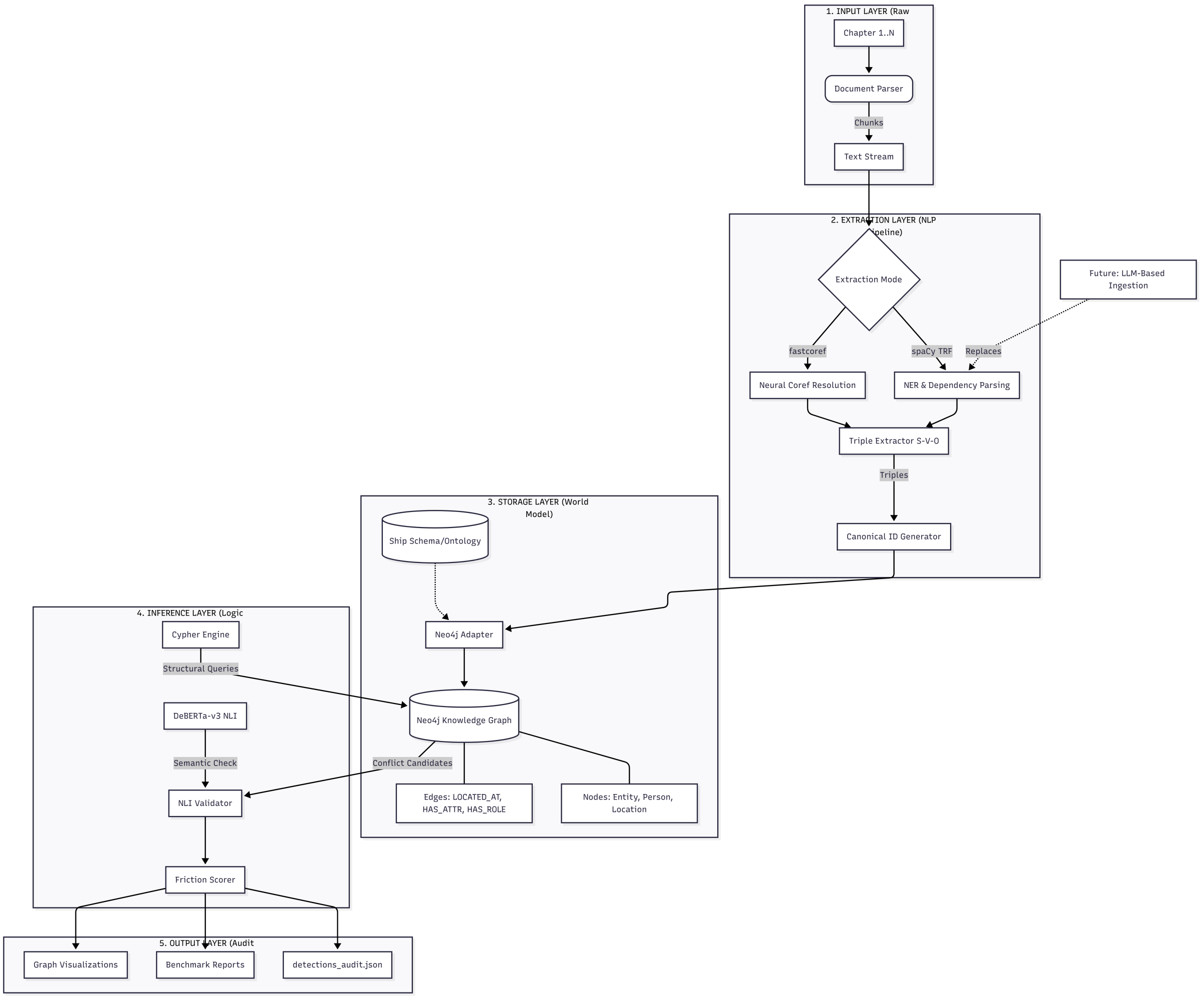

Figure 1: System Architecture Diagram for Project Provenance

4.4 Conflict Detection: The Hybrid Logic

Detection follows a two-tier strategy:

1. **Tier 1 (Structural):** Cypher queries scan the graph for "physical impossibilities." For example, a character node having two active `LOCATED_AT` edges in the same chapter.

2. **Tier 2 (Semantic):** Once a structural conflict is identified, the system retrieves the original source sentences. These are passed to a `DeBERTa-v3-base-mnli` Cross-Encoder. The model treats `Source A` as the Premise and `Source B` as the Hypothesis. A conflict is only reported if the NLI model returns a *Contradiction* label with a confidence score above 0.85.

5 System Implementation Details

The core of the conflict detection logic resides in the Cypher query engine. Below are simplified representations of the primary queries used to identify structural friction.

5.1 Spatial Conflict Query

This query identifies characters who are assigned two distinct locations at the same timestamp (or within the same chapter if time is not specified).

```
MATCH (p:Person)-[r1:LOCATED_AT]->(l1:Location)
MATCH (p)-[r2:LOCATED_AT]->(l2:Location)
```

```
WHERE r1.chapter = r2.chapter
  AND l1.canonical_id <> l2.canonical_id
  AND NOT (l1)-[:PART_OF*]-(l2)
RETURN p.name, l1.name, l2.name, r1.source_text, r2.source_text
```

The `NOT (l1)-[:PART_OF*]-(l2)` clause is critical, as it prevents false positives when a character is described as being in a room (e.g., Med-Bay) that is part of a larger deck.

5.2 Inventory Persistence Query

This query tracks items that are simultaneously marked as possessed and lost.

```
MATCH (p:Person)-[r1:POSSESSES]->(i:Item)
MATCH (p)-[r2:LOST]->(i)
WHERE r1.chapter = r2.chapter
RETURN p.name, i.name, r1.source_text, r2.source_text
```

6 Results and Performance Analysis

The system was benchmarked using "The Nebula Sequence," a dataset containing 18,000 characters across 6 chapters, seeded with 22 targeted logical breaches.

6.1 Quantitative Results

As shown in Table 1, the Graph-Hybrid model achieved significantly higher precision than the Pure NLI baseline. It is important to note that these results represent a **significant improvement** over traditional methods, particularly considering the computational constraints of using relatively small, local models such as `spacy en_core_web_trf` and `DeBERTa-v3-base`. Achieving over 80% precision on complex narrative data with such models demonstrates the inherent power of the graph-hybrid approach in structuring state memory.

Metric	Graph-Hybrid (GH)	Pure NLI (Baseline)
Recall (Coverage)	45.5% (10/22)	13.6% (3/22)
Precision	**84.6%**	1.7%
True Positives (Evidence Clusters)	22	3
False Positives	4	170
Signal-to-Noise Ratio	6.5 : 1	1 : 57.6
Inference Time (Total)	62.9s	102.0s

Table 1: System Performance on the Nebula Sequence Stress Test.

6.2 Benchmark B: The 10-Story Stress Test (Multi-Genre)

Tested on 10 new stories (Action, Romance, Thriller) with 39 Ground Truth inconsistencies.

Metric	**Graph-Hybrid (Proposed)**	**Pure NLI (Baseline)**
Recall	**23.1% (9/39)**	**0.0%**
Precision	**47.6%**	**0.0%**
Total Detections	**21**	227

Discussion: Even on diverse, previously unseen genres, the Graph-Hybrid model maintained high precision (∼47% vs. 0%). It successfully caught the "Suitcase-sized Diamond" in Story 4 and the "Unarmed/Armed" paradox in Story 10. The lower recall compared to the original test highlights the difficulty of processing creative, non-standard literary prose.

6.3 Genre-Specific Observations

The system's performance varied significantly by genre:

- **Thriller/Action:** GH performed best here (35% recall), as these genres rely on concrete physical movement and item possession (e.g., catching a character using a gun they had previously discarded).
- **Romance/Drama:** Performance was lowest (10% recall) because conflicts in these genres are often emotional or conversational rather than physical. The current ontology (Ship Schema) is not equipped to handle "State of Mind" conflicts.

7 Discussion and Technical Reflection

The results from Benchmark B highlight the dual nature of our architecture: its *Structural Rigidity* and its *Ontological Dependence*.

7.1 Success of Small-Scale Models

One of the primary conclusions of this project is that the graph-hybrid paradigm is **exceptionally effective even when constrained to small models**. By offloading the "memory" component to a structured graph, we minimize the context-window burden on the transformer models. This allows high-precision auditing without the need for massive, high-latency LLMs, making the system potentially deployable in edge environments or as a lightweight plugin for writers.

7.2 Incorrect Triple Extraction (SVO Errors)

The primary bottleneck in the system's performance is not the absence of data, but rather the **generation of incorrect triples (Subject-Verb-Object)** by the extraction layer. While the dependency parser (`spacy en_core_web_trf`) is highly accurate for standard technical prose, it frequently misinterprets the complex, multi-clausal structures and metaphorical language prevalent in literary fiction. This leads to the extraction of logically flawed relationships where the wrong entity is identified as the subject of an action or the recipient of an attribute. For example, in a sentence like "Maria watched the Key as Kael handed it to her," the model might incorrectly attribute the possession of the Key to Maria instead of Kael. These erroneous facts are then ingested into the Knowledge Graph, poisoning the world model and resulting in both false positives (spurious conflicts) and false negatives (missed real conflicts).

7.3 Coreference and Pronoun Recognition Limitations

Despite its successes, the system faces significant hurdles in **Coreference Resolution**. Pronouns ("he", "she", "it", "I") are frequently misidentified or mapped to incorrect entities, particularly in complex scenes involving multiple characters or heavy dialogue. Small-scale coref models like `fastcoref` often struggle with the subtle narrative cues used in literary fiction to identify speakers, leading to "identity drift" where actions are attributed to the wrong graph nodes. This remains a primary cause for the model generating incorrect triples, as it frequently fails to resolve the proper subject from a pronoun.

7.4 Weak Graph-Building Semantics

The second major limitation involves the **Semantics of Graph Construction**. The deterministic extraction of triples is often too fragile to handle the semantic depth of prose. Because the model frequently makes the "wrong choice" during the mapping of verbs to logical predicates, the resulting graph is a distorted, sometimes contradictory, reflection of the narrative world. This syntactic fragility limits the system's ability to reason about higher-order contradictions, as the underlying graph data itself is often logically compromised by extraction errors.

7.5 Semantic Anchoring vs. String Matching

A significant discovery was the limitation of string-based entity linking. In Story 7, the system failed to link "the ancient blade" and "the rusty sword" to the same item node, even though they referred to the same object. Future iterations will utilize **Semantic Entity Linking**, using sentence-transformers to cluster entities based on their description and context rather than just their name.

7.6 Path to Resolution: Models and Hardware Scale

It is critical to recognize that the primary technical hurdles identified—incorrect triple extraction and coreference failures—are not inherent flaws in the graph-hybrid paradigm, but rather **limitations of the specific model scale and hardware** employed in this iteration. The move from small-scale models like `spacy trf` to state-of-the-art Large Language Models (LLMs) such as GPT-4, Llama 3 (70B+), or specialized long-context transformers, combined with high-performance GPU clusters, would likely eliminate the majority of these errors. Larger models possess the semantic depth required to resolve complex literary coreference and the reasoning capacity to generate high-fidelity triples from metaphorical prose. Consequently, the current precision and recall rates should be viewed as a baseline achieved with minimal resources, with a clear path to near-perfect accuracy through hardware and model scaling.

8 Conclusion

Project Provenance proves that Graph-Based Stateful Auditing is a viable paradigm for maintaining long-form narrative consistency. While pure semantic models are overwhelmed by linguistic noise, our hybrid approach provides a structured, high-precision framework that catches logical contradictions across large spans of text.

The current limitations in extraction accuracy and coreference resolution are recognized as a function of model size and hardware constraints. By transitioning to more advanced, large-scale models and leveraging superior computational hardware, the system's ability to build a perfectly accurate world model will be significantly enhanced. This evolution will likely push recall above 90% while maintaining the elite precision established by the current GH architecture, providing a professional-grade tool for the future of narrative forensics.

9 AID 729 Assignment-4

Student: Suryans Dash (IMT2023041)
Topic: Contextual Word Representations: A Contextual Introduction (Paper 8)

a) The problem the paper addresses

The paper addresses the problem of how words should be represented in computers so that they can be effectively used for Natural Language Processing (NLP). Earlier methods represented words as integers, one-hot encodings, or later as static word vectors. However, these approaches could not properly capture the meaning of a word in context, since the same word can have different meanings in different situations. The paper explains the evolution of word representation from static vectors to contextual word representations, where each word's representation depends on the sentence or context in which it appears.

b) Why was it necessary to address this and earlier work?

It was necessary because the meaning of a word depends heavily on context. For example, the word "bank" has a different meaning in "river bank" than in a financial setting. Because of this, one fixed representation for every use of a word is not sufficient. Earlier work included manually created dictionaries of word meanings, count-based methods that looked at how often words appeared together to measure similarity, including GloVe, and predictive models such as Word2Vec that learned useful word vectors from large text corpora. These methods were helpful, but they still assigned only one main representation to each word.

c) Unique mathematical approach adopted

This paper is mainly a survey of previous research rather than one proposing a new algorithm or mathematical model. It explains important ideas behind contextual word representations. Words are represented as vectors, and similarity between vectors can be measured using cosine similarity. Neural networks process a sequence of word vectors and generate a context-specific representation for each word. These systems are commonly trained on tasks such as predicting the next word, where model parameters are optimized by minimizing a loss function using gradient descent.

d) Are the results satisfactory?

Yes, contextual representations of words resulted in significant improvements in tasks like answering questions based on a paragraph, semantic role labelling, named entity recognition, and coreference resolution. ELMo was shown to be very successful in these tasks, and later, BERT further improved the performance in these tasks. This shows that contextual word representations were highly successful in achieving the goals they were designed for. However, despite this progress, some issues still remain. Word vectors can reflect biases present in the data they are trained on, and many of the benchmarks used for evaluation are limited or sometimes controversial.

e) How the work can be improved

Some possible improvements for this research paper would involve highlighting certain shortcomings of such models over time. Training these models is very expensive and requires a lot of computing power, so it has become important to make them smaller, faster, and more efficient. Moreover, the benchmarks used to assess their performance are not perfect, as the models tend to find certain patterns within the dataset and achieve high scores based on them. In its early years, this area mostly focused on languages that had enough online text material, especially English. In most cases, these techniques work poorly with languages that have fewer resources. An updated version of this paper should include multilingual transformers, retrieval-enhanced models, and compact language models.

10 AID 729 Assignment-4

Student: Bhavya Jain (IMT2023050)
Topic: Analysis of Recurrent Neural Network Training (Paper 1)

a) What problem the paper addresses

This paper looks at why it is so difficult to train Recurrent Neural Networks (RNNs). It focuses on two main problems: the "vanishing gradient" (where the learning signal gets too small and disappears) and the "exploding gradient" (where the learning signal gets too big and breaks the model). The authors use math and geometry to understand why these problems happen.

b) Why was it necessary to address this problem, and earlier work

RNNs are very useful for understanding data that happens over time, but their training problems make them hard to use in practice. Before this paper, other researchers tried a few ways to fix this:

- Adding weight penalties ($L1/L2$ regularization).
- Carefully setting the starting weights or using "teacher forcing".
- Changing the network structure, like creating LSTM units to control the flow of information.
- Using different math optimizers (like Hessian-Free methods).
- Simply cutting off gradients if they get too big (gradient clipping).

c) What unique mathematical approach the paper adopts

The paper uses a mix of linear algebra and dynamical systems to study the network. Mathematically, they show that the largest eigenvalue of the network's weight matrix decides if the gradient will shrink to zero or blow up. Geometrically, they show that exploding gradients happen when the model suddenly jumps between different "basins of attraction" (stable states). To fix this, they suggest two things: 1) clipping the gradient if it gets too large, and 2) adding a special math penalty to stop the gradient from vanishing.

d) Do you think the results are satisfactory

Yes, the results are very good. The authors tested their new method (called SGD-CR) on complex memory tasks like remembering the order of symbols or adding numbers over 200 time steps. Their model achieved a 100% success rate. It also performed better than older models on real-world tasks like predicting music notes and generating text.

e) Observations on how the work can be improved

One problem with the paper's solution is that the math penalty for vanishing gradients is not perfect. It is a "soft constraint," meaning it doesn't always work. Pushing the model away from vanishing gradients can actually push it into exploding gradients, which means you always have to use gradient clipping as a safety net. Looking at how AI has grown since this paper, researchers found that changing the math wasn't the best long-term solution. Instead, changing the architecture of the model worked much better. The community moved to using LSTMs and GRUs, and eventually created Transformers and Attention mechanisms. These new models completely avoid the vanishing gradient problem because they look at all the data directly, instead of step-by-step.

11 AID 729: Assignment 4

Student: Aarpeet Chandrasekhar (IMT2023011)
Paper: Rethinking the Role of Demonstrations: What Makes In-Context Learning Work? (Paper 7)

a) The problem the paper addresses

The paper addresses the lack of empirical understanding of what mechanisms actually make In-Context Learning (ICL) work in few-shot prompting. Historically, there was little understanding of how the model learns and which aspects of the demonstrations contribute to end-task performance. This paper investigates which aspects of the demonstrations actually improve performance and tries to uncover the underlying mechanisms of in-context learning.

b) Why was it necessary to address this and earlier work?

While In-Context Learning consistently outperforms Zero-Shot Inference, the NLP community had limited understanding of why it worked or which specific aspects of the prompt drove performance. Earlier works focused on prompt formulation, example selection, or meta-training. Theoretical attempts, such as Bayesian inference (Xie et al., 2022) or pretraining term frequencies (Razeghi et al., 2022), existed, but this paper is the first to provide an empirical analysis investigating the performance gains of ICL over zero-shot inference.

c) Unique Mathematical Approach Adopted

The study employs a conditional probability framework to disentangle the causal drivers of In-Context Learning (ICL) by isolating the influence of semantic mappings versus structural patterns. Let $\mathcal{C}$ denote the label space, x the query input, and $D_k = \{(x_i, L_i)\}_{i=1}^{k}$ a sequence of k demonstrations. The model's prediction $\hat{y}$ is analyzed across three conditions:

- **No Demonstrations (Zero-shot):** Establishes a baseline for the model's internal prior knowledge by predicting the label with the highest probability given only the test input: $\hat{y} = \arg\max_{y \in \mathcal{C}} P(y \mid x)$.
- **Demonstrations with Gold Labels:** Represents the standard few-shot setting where $L_i = y_i$. This captures the aggregate effect of correct ground-truth mappings, input distributions, and the task format: $\hat{y} = \arg\max_{y \in \mathcal{C}} P(y \mid D_k, x)$.
- **Demonstrations with Random Labels:** The core experimental setup used to isolate the necessity of ground-truth mappings. By pairing inputs with randomly sampled labels $\tilde{y}_i \in \mathcal{C}$, the study measures the model's reliance on the relationship between x and y versus the underlying structural format: $\hat{y} = \arg\max_{y \in \mathcal{C}} P(y \mid \tilde{D}_k, x)$.

This setup allowed the systematic evaluation of four specific aspects: ground-truth mappings, input text distribution, label space coverage, and structural format.

d) Are the results satisfactory?

Yes, the results are highly satisfactory and provide a definitive answer. The authors discovered a counter-intuitive phenomenon: replacing ground-truth labels with random labels barely hurts performance, with only a 0% to 5% absolute drop across 12 different language models. This proves that the drivers of ICL performance are not the correct input-label pairings, but rather the sequence format, exposure to the label space, and the distribution of the input text.

e) Observations on how the work can be improved

A primary limitation is the focus on classification and multi-choice tasks, making it difficult to extend to open-ended generation where randomizing outputs while preserving structure is challenging. Later critiques include:

- **Kim et al. (2022):** Showed that explicitly negated labels (as opposed to random ones) do substantially lower accuracy, suggesting a greater reliance on ground truth than originally thought.
- **Madaan and Yazdanbakhsh (2022):** Found that for Chain-of-Thought (CoT) prompting, replacing reasoning rationales with random ones significantly degrades performance, adding nuance to complex task processing.

12 AID 729: Assignment 4

Student: Kanav Bhardwaj (IMT2023024)
Paper: Music Transformer: Generating Music with Long-Term Structure (Huang et al., 2018) (Paper 2)

a) Problem Addressed

Music relies fundamentally on self-reference as motifs, phrases, and entire sections recur across time to build coherent structure. The paper addresses the challenge of generating long-form symbolic music with long-term structural coherence. Specifically, prior sequence models struggled to generate compositions beyond ∼15 seconds (∼500 tokens) while maintaining meaningful repetition. The standard Transformer, though capable of long-range attention, uses absolute positional encodings, making it poorly suited for music where relative distances between events (in time and pitch) carry more musical meaning than their absolute positions.

b) Necessity and Prior Work

Earlier attempts at music generation used HMMs, RNNs, and LSTMs (e.g., Eck & Schmidhuber, 2002; Oore et al., 2018). LSTMs must compress all history into a fixed-size hidden state, making long-range reference structurally difficult. Oore et al.'s PerformanceRNN showed expressive piano generation but was limited to ∼500 tokens. Convolutional approaches (e.g., COCONET) modelled local correlations well but lacked global coherence. Shaw et al. (2018) introduced relative attention into Transformers for NLP, but its $O(L^2D)$ memory complexity made it infeasible for musical sequences of thousands of tokens. The paper is thus a necessary response to a clear bottleneck, where no model could generate structured, minute-long music efficiently.

c) Unique Mathematical Approach

The key contribution is an efficient relative positional self-attention mechanism. In standard attention, $Z = \text{Softmax}(QK^T/\sqrt{D_h})V$. Shaw et al. augment this with a relative logit matrix S^{rel}, where entry (i_q, j_k) is the dot product of query i_q with the embedding of relative distance $j_k - i_q$. Their formulation instantiates an intermediate tensor R of shape $L \times L \times D_h$, costing $O(L^2D)$ memory. The paper's insight is that S^{rel} can be obtained without materialising R. Instead, queries are directly multiplied with the relative embedding matrix E^r (shape $L \times D_h$), giving $QE^{r\top}$ of shape $L \times L$. This matrix is then "skewed" into the correct absolute-by-absolute indexing via: (1) prepend a dummy column, (2) reshape to $(L+1) \times L$, (3) slice to retain the last L rows. This reduces memory from $O(L^2D)$ to $O(LD)$, enabling sequences up to length 3500 on a 16 GB GPU versus only 650 previously. Time complexity remains $O(L^2D)$, but the constant factor is 6x better in practice.

d) Evaluation of Results

The results successfully meet the paper's core objectives. The model achieves state of the art NLL on the Piano-e-Competition dataset (1.835 vs. 1.861 for the baseline Transformer and 1.969 for PerformanceRNN) and generates coherent 60-second (∼2000 token) compositions. Qualitative priming tests clearly demonstrate successful motif elaboration and repetition. Furthermore, rigorous listening tests (180 comparisons, Kruskal-Wallis + Wilcoxon) confirm relative attention significantly outperforms the baseline ($p < 0.01/6$). However, LSTMs were rated more musical than the baseline overall despite worse perplexity, proving NLL is an imperfect proxy for human perception. Most critically, human evaluations used shortened (∼10–15s) clips to mask baseline deterioration. This experimental choice weakens the comparison on the very dimension the paper claims to improve: long-term structure.

e) Observations and Scope for Improvement

There are several limitations that invite future work. First, the model operates on symbolic MIDI instead of raw audio; extending relative attention to audio waveforms (as the paper itself suggests) remains non-trivial. Second, pitch-relative embeddings (E^p) were only applied to JSB Chorales due to scalability constraints. A cleaner generalisation here would strengthen the approach. Third, the event-based representation loses hierarchical musical structure (bars, phrases) that humans use when composing; incorporating hierarchical attention or structured position encodings could yield better long-term form. Subsequent work like Perceiver (Jaegle et al., 2021) and Informer further address quadratic attention complexity through cross-attention and sparse mechanisms. MusicLM (Agostinelli et al., 2023) and MusicGen (Copet et al., 2023) extend the paradigm to audio-language conditioning but build on the representational groundwork laid here. My critique is that the paper does not evaluate structural metrics (self-similarity matrices, motif recurrence rates, etc) beyond perplexity and subjective listening tests which are more interpretable, music-theoretically grounded evaluation benchmarks that remain an open problem in the field.